

Verified Completion for Agentic AI

Why “the agent ran” is not the same as “the outcome is proven done”

BRAIZ Works LLC · WP-01 · Version 1.0.0 · September 24, 2026

Public-safe research. Principles and control outcomes only; proprietary implementation remains protected.

Executive summary

Agentic systems can execute impressive sequences of work while leaving enterprises uncertain about whether the intended outcome was actually achieved, whether the right authority existed, whether evidence is current, and whether the result can survive independent review. This paper defines verified completion as a public-safe operating discipline that binds outcome, authority, evidence, lifecycle, recovery, and review without disclosing proprietary control-plane implementation.

The completion problem

AI systems are increasingly capable of executing multi-step work, but execution traces and plausible outputs do not by themselves prove outcome closure. Enterprise use introduces additional questions: Was the intended result explicit? Did the system have authority for every consequential action? Are the facts and files current? Can a reviewer reproduce the evidence? Is there a recovery path if the result is wrong?

The distinction matters because organizational AI use is broad while enterprise-wide scaling remains much less common. The 2026 AI Index and McKinsey both describe widespread adoption alongside continuing scaling challenges. [1][2] A system that optimizes for activity can therefore look productive while still leaving governance, assurance, and operational risk unresolved.

A public definition of verified completion

Verified completion is the state in which an authorized outcome is not merely attempted but supported by current, attributable, reproducible evidence across its material obligations. The concept is narrower than “perfect” and stronger than “done.” It asks whether the exact in-scope result has enough proof to justify its lifecycle state.

A useful public abstraction is:

Verified completion = explicit outcome + valid authority + current evidence + lifecycle truth + recovery + independent challenge.

This aligns with NIST’s emphasis on governing, mapping context, measuring, managing, documenting tests, and using independent review where appropriate. [3][4][5]

Six conditions for closure

Condition	Enterprise question
Outcome	What exact result must exist when the work is done?
Authority	Was every consequential action authorized for this exact scope?
Evidence	Can material claims be traced to current, attributable proof?
Lifecycle	Is the reported state exactly what has actually been achieved?

Condition	Enterprise question
Recovery	Can the system return to a known-good state and prove the restore?
Review	Can a structurally separate reviewer challenge the result without relying on the builder's conclusion?

None of these conditions requires publishing an internal implementation. They are outcome-level controls that enterprises can ask any system to satisfy.

Lifecycle truth is a control, not a label

A common failure mode is to collapse different states into one generic “complete.” A candidate is not independently reviewed. An independently reviewed artifact is not automatically owner accepted. A frozen artifact is not automatically deployed. A successful deployment is not automatically observed in operation.

Explicit lifecycle states prevent a system from using downstream language to hide an upstream gap. They also make automation safer: the next action can be derived from the first unmet gate rather than from a vague sense of progress.

Evidence must be bound to the subject

Evidence becomes weak when it is detached from the exact thing it is supposed to prove. A test from an earlier version, a screenshot without a subject identity, a manifest that was never independently recomputed, or a “PASS” flag copied forward after a material change can all create false confidence.

Evidence-first operation therefore requires identity, currentness, provenance, role, and recomputability. The evidence should answer not only “what happened?” but also “to which exact subject, under which authority, and can another reviewer reproduce it?”

Recovery is part of completion

A backup is not the same as recovery. A stronger recovery chain establishes that a backup exists, its integrity is verified, a restore is actually tested, and the restored state is verified. The point is not ceremony; it is to ensure that a claimed recovery path works before it is needed under pressure.

Independent review without blueprint disclosure

The builder should be allowed to test aggressively and repair. Independent review should remain separate and should not silently repair the subject it is judging. This separation reduces the risk that the same assumptions, shortcuts, or blind spots drive both production and approval.

Public documentation can describe this discipline without disclosing proprietary prompts, routing logic, exact thresholds, private attack sets, or internal control-plane state. The public value is the assurance principle; the protected value is the implementation.

What enterprises can ask for now

Enterprises do not need to adopt a specific architecture to demand stronger completion semantics. They can require exact outcome contracts, authority boundaries, evidence bindings, explicit lifecycle states, tested recovery, and independent review for consequential work. These controls are compatible with different models, agent frameworks, orchestration systems, and deployment environments.

The central shift is simple: move the optimization target from “the agent did something useful” to “the authorized outcome is proven complete within a defined evidence boundary.”

Limitations

This paper presents an operating concept, not a claim that any system can eliminate all defects or guarantee perfect outcomes. “Verified” is always bounded by the authenticated scope, current evidence, test coverage, and known uncertainty. The paper intentionally omits private UBuildOS control-plane implementation, prompt logic, routing recipes, thresholds, internal evidence schemas, and proprietary evaluation corpora.

References

- [1] Stanford Institute for Human-Centered Artificial Intelligence. *The 2026 AI Index Report — Economy*. 2026. <https://hai.stanford.edu/ai-index/2026-ai-index-report/economy>
- [2] McKinsey & Company. *AI at work but not at scale*. December 10, 2025. <https://www.mckinsey.com/featured-insights/charts/ai-at-work-but-not-at-scale>
- [3] Tabassi, E. *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*. NIST AI 100-1. National Institute of Standards and Technology, 2023. <https://doi.org/10.6028/NIST.AI.100-1>
- [4] Autio, C., Schwartz, R., Dunietz, J., Jain, S., Stanley, M., Tabassi, E., Hall, P., Roberts, K. *Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile*. NIST AI 600-1, 2024; NIST page updated April 8, 2026. <https://doi.org/10.6028/NIST.AI.600-1>
- [5] National Institute of Standards and Technology. *NIST AI RMF Playbook*. <https://www.nist.gov/itl/ai-risk-management-framework/nist-ai-rmf-playbook>
- [6] BRAIZ Works LLC. *UBuildOS Verified Completion™ Thesis v1.0.5*. September 15, 2026. DOI: <https://doi.org/10.5281/zenodo.22775537> ; public repository: <https://github.com/BRAIZ-Works/ubuildos-verified-completion-thesis>
- [7] BRAIZ Works LLC. *The Enterprise Execution Gap*. BRAIZ Works Research WP-02 v1.0.0. September 23, 2026.

Source-currentness note

External references [1]-[5] are carried forward from the independently reviewed and publicly released WP-02 v1.0.0 canary. This batch does not claim a new live-web currentness check. Time-sensitive statistics are used only where explicitly cited and dated. First-party references [6]-[7] are BRAIZ Works publications.

Rights and disclosure boundary

© 2026 BRAIZ Works LLC. All rights reserved.

This public artifact intentionally excludes private prompts, proprietary routing/controller logic, internal thresholds, private attack sets, protected evidence schemas, credentials, customer data, and reconstruction-sensitive implementation details.